

CLASS TEST - 3

- ① Show that the field tensor $F_{\mu\nu}$ is invariant under gauge transformation.

Ans:

Gauge transformation:

$$A'^{\mu} = A^{\mu} + \nabla^{\mu} f(t, r)$$

$$F_{\mu\nu} = \partial_{\mu} A_{\nu} - \partial_{\nu} A_{\mu}$$

$$\begin{aligned} \therefore F'_{\mu\nu} &= \partial_{\mu} A'_{\nu} - \partial_{\nu} A'_{\mu} \\ &= \partial_{\mu} [A_{\nu} + \nabla_{\nu} f] - \partial_{\nu} [A_{\mu} + \nabla_{\mu} f] \\ &= (\partial_{\mu} A_{\nu} - \partial_{\nu} A_{\mu}) + (\partial_{\mu} \nabla_{\nu} f - \partial_{\nu} \nabla_{\mu} f) \\ &= F_{\mu\nu} \quad \text{[partial derivatives commute]} \end{aligned}$$

$$\therefore \boxed{F'_{\mu\nu} = F_{\mu\nu}} \quad \text{(Proved)}$$

- ② An observer O in her frame sees a charge density ρ_0 but no current i.e., $\vec{J} = 0$. By using the Lorentz transformation, find the charge & current density as seen by another observer O' , who is moving with a speed v along the x -direction wrt O .

Ans:

$$\Lambda = \begin{bmatrix} \gamma & -\frac{\gamma v}{c} & 0 & 0 \\ -\frac{\gamma v}{c} & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$\mu=0$, rest frame $\vec{J}^{\mu} = (c g_0, 0, 0, 0)$

μ' rest frame $\vec{J}'^{\mu} = (c g', J_x', J_y', J_z')$

$\mu=0$ "Transform": $J'^{\mu} = \Lambda^{\mu}_{\nu} J^{\nu}$

$\mu=0$:

$$J'^0 = \Lambda^0_{\nu} J^{\nu} = \Lambda^0_0 J^0 + \Lambda^0_1 J^1 \quad [J^1 \neq 0]$$

$$\rightarrow c g' = \gamma c g_0 = J'^0 \quad \text{--- (i)}$$

$$\rightarrow g' = \gamma g_0$$

$\mu=1$:

$$\begin{aligned} J''^1 &= \Lambda^1_{\nu} J^{\nu} \\ &= \Lambda^1_0 J^0 + \Lambda^1_1 J^1 \\ &= -\frac{\gamma v}{c} \cdot c g_0 \end{aligned}$$

$$\rightarrow J''^1 = -\gamma v g_0 \quad \text{--- (ii)}$$

$\mu=2$: $J'^2 = \Lambda^2_{\nu} J^{\nu} = \Lambda^2_2 J^2 = 0 \quad \text{--- (iii)}$

$\mu=3$: $J'^3 = \Lambda^3_{\nu} J^{\nu} = \Lambda^3_3 J^3 = 0 \quad \text{--- (iv)}$

From (i) - (iv):

$$J'^{\mu} = (\gamma c g_0, -\gamma v g_0, 0, 0) \quad \text{(Ans)}$$

$$g' = \gamma g_0 \quad \text{(Ans)}$$